

GRAND CANYON NATIONAL PARK, AZ, USA

WMO: 723783

Lat: 35.946N

Lon: 112.155W

Elev: 6606

StdP: 11.51

Time Zone: -7.00 (W07)

Period: 94-19

WBAN: 03195

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-1.2	5.1	-10.6	4.0	17.8	-6.2	5.0	23.5	25.5	40.5	22.4	37.6	1.7	50	0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	34.2	89.8	56.2	87.6	55.8	84.8	55.3	62.1	74.5	61.0	74.1	60.0	74.1	10.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.8	95.1	64.6	57.1	89.1	64.3	55.3	83.4	64.0	31.1	74.0	30.2	74.0	29.3	73.8	70.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.6	20.7	18.2	DB	-12.4	93.5	7.8	2.3	-18.0	95.2	-22.6	96.5	-27.0	97.8	-32.7	99.5	(4)
(5)			WB	-12.5	64.9	7.8	1.8	-18.1	66.2	-22.7	67.3	-27.1	68.3	-32.7	69.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.8	29.7	31.9	38.2	43.2	51.1	60.7	67.4	65.6	58.6	47.1	37.4	29.3		
	DBStd	14.76	8.05	8.14	6.69	6.30	6.33	5.35	3.49	3.16	4.91	6.43	7.26	8.06		
	HDD50	2938	629	506	366	215	65	3	0	0	6	127	379	642		
	HDD65	6813	1094	926	830	654	432	144	17	31	197	553	828	1107		
	CDD50	1756	0	0	1	10	98	323	539	483	263	38	1	0		
	CDD65	160	0	0	0	0	0	15	91	50	4	0	0	0		
	CDH74	6363	0	0	0	8	282	1853	2215	1478	498	29	0	0		
	CDH80	2065	0	0	0	0	44	661	854	439	66	1	0	0		
	WSAvg	6.2	5.8	6.4	6.7	7.7	7.5	7.4	5.3	4.8	5.6	5.9	5.8	5.7		
	PrecAvg	12.5	1.0	1.0	0.9	0.8	0.5	0.4	1.5	2.0	1.3	1.1	0.6	1.1		
(15) Precipitation	PrecMax	20.4	4.3	3.1	2.0	1.9	1.7	2.0	2.9	4.4	3.9	4.4	2.7	2.6		
	PrecMin	7.1	0.0	0.1	0.0	0.1	0.0	0.0	0.0	0.4	0.1	0.0	0.0	0.2		
	PrecStd	3.9	1.1	0.7	0.6	0.5	0.5	0.5	0.7	1.1	1.0	1.4	0.6	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.6	63.3	70.1	75.1	84.3	92.3	92.9	90.1	84.4	77.4	69.4	59.1		
		MCWB	39.6	42.0	45.4	47.3	52.8	55.4	57.6	56.6	54.9	50.8	45.9	40.8		
	2%	DB	54.4	58.4	65.5	71.4	80.6	89.3	90.2	87.5	81.8	73.4	64.4	54.5		
		MCWB	38.5	40.5	43.4	46.2	51.0	54.5	57.1	57.2	54.1	49.0	43.4	38.8		
	5%	DB	51.1	54.1	61.8	67.8	77.1	86.7	87.9	84.7	79.5	70.9	61.3	50.5		
		MCWB	36.9	38.3	41.8	44.5	48.9	53.4	56.7	57.0	53.6	47.9	42.1	36.9		
	10%	DB	46.3	49.6	57.8	64.2	73.4	83.9	85.1	82.1	76.9	67.4	56.9	46.0		
		MCWB	34.9	36.7	40.2	43.1	47.8	52.6	56.5	56.9	53.4	46.7	40.5	34.9		
	0.4%	WB	43.1	44.3	46.9	49.2	54.6	59.0	63.8	63.4	61.5	55.5	47.4	43.5		
		MCDB	52.7	56.0	65.2	71.7	79.2	80.2	75.7	75.9	71.5	65.8	63.4	54.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	40.2	41.9	44.5	47.2	52.2	56.9	62.3	62.1	59.5	52.6	45.2	41.2		
		MCDB	49.9	54.0	62.4	67.7	75.4	81.4	74.9	74.7	71.9	66.4	60.0	50.1		
	5%	WB	38.0	39.8	42.7	45.3	50.6	55.3	61.3	61.0	57.9	50.5	43.4	38.6		
		MCDB	46.9	51.0	59.5	65.3	73.2	80.7	74.3	74.0	70.7	65.2	58.0	47.0		
	10%	WB	36.1	37.6	41.0	43.6	49.0	53.8	60.2	60.0	56.4	48.4	41.5	36.2		
		MCDB	44.9	47.2	56.5	62.9	71.0	79.9	74.5	73.8	70.1	63.6	55.2	44.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	28.6	28.2	31.7	33.8	37.3	42.3	34.2	32.1	34.0	34.3	32.3	28.4		
		MCDBR	35.8	37.7	40.8	42.9	45.2	47.4	41.1	38.7	39.3	42.0	41.0	36.3		
		MCWBR	21.7	21.7	21.4	21.4	21.6	21.7	16.2	15.0	17.4	21.5	23.2	22.5		
		MCDBR	29.1	30.9	36.9	38.7	40.2	41.5	27.6	27.9	28.5	33.2	35.7	29.3		
	5% WB	MCWBR	18.6	18.6	19.9	19.8	19.4	19.3	10.3	11.6	13.6	17.8	21.1	19.1		
(40) Clear-Sky Solar Irradiance	taub		0.236	0.241	0.257	0.272	0.282	0.286	0.331	0.339	0.289	0.271	0.241	0.234		
	taud		2.642	2.643	2.586	2.525	2.547	2.526	2.467	2.428	2.590	2.591	2.642	2.654		
	Ebn at Noon		316	325	325	321	317	314	299	295	309	308	311	311		
	Edh at Noon		22	25	29	33	32	33	35	36	29	26	22	20		
(44) All-Sky Solar Radiation	RadAvg		976	1262	1748	2185	2445	2649	2182	2007	1863	1461	1105	862		
	RadStd		85	115	117	108	174	138	139	112	81	111	53	65		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+61
Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+131	N/A

Nomenclature: See separate page